

AP 2016 Class 12 Chemistry

Paper-II

Time:3 Hours

Max.Marks:60

Note: Read the following instructions carefully:

- 1) Answer all the questions of Section-‘A’. Answer any six questions from Section-‘B’ and any two questions from Section-‘C’.
- 2) In Section-‘A’, Questions from Sr. Nos. 1 to 10 are of “very short answer type”. Each question carries two marks. Every answer may be limited to two to three sentences. Answer all the questions at one place in the same order.
- 3) In Section-‘B’, Questions from Sr. Nos. 11 to 18 are of “Short answer type”. Each questions carries four marks. Every answer may be limited to 75 words.
- 4) In Section-‘C’, Questions from Sr.Nos.19 to 21 are of” Long answer type”. Each questions carries eight marks. Every answer may be limited to 300 words.
- 5) Draw labelled diagram, wherever necessary for questions in section-‘B’ and ‘C’.

SECTION- A

10 × 2=20

Note: Answer all the questions:

1. What is PHBV? How is it useful to man?

Ans:- PHBV is poly β – Hydroxy butyrate Co – β Hydroxy valerate. It is a copolymer, formed due to condensation of 3 – hydroxy butanoic acid and 3 – hydroxy pentatonic acid. This biodegradable polymer is used for making capsules and orthopaedic devices.

2. Write the names of monomers of the following polymers:

(a) Bakelite:-

Ans:- Phenol, Formaldehyde

(b) Terylene:-

Ans:- Terephthalic acid, Ethylene Glycol

3. Define osmotic pressure.

Ans:- Osmotic pressure is the minimum pressure which needs to be applied to a solution to prevent the inward flow of its pure solvent across a semipermeable membrane. ... Solvent molecules pass preferentially through the membrane from the low-concentration solution to the solution with higher solute concentration.

4. State Faraday's first law of electrolysis.

Ans:- Faraday's First Law of Electrolysis states that the chemical deposition due to the flow of current through an electrolyte is

directly proportional to the quantity of electricity (coulombs) passed through it.

i.e. mass of chemical deposition:

$$m \propto \text{Quantity of electricity, } Q \Rightarrow m = Z \cdot Q$$

Where, Z is a constant of proportionality and is known as electrochemical equivalent of the substance.

If we put $Q = 1$ coulombs in the above equation, we will get $Z = m$ which implies that electrochemical equivalent of any substance is the amount of the substance deposited on the passing of 1 coulomb through its solution. This constant of the passing of electrochemical equivalent is generally expressed in terms of milligrams per coulomb or kilogram per coulomb.

5. What is poling?

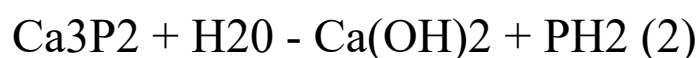
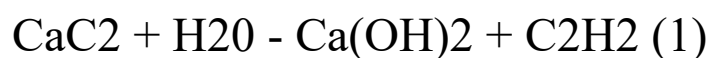
Ans:- **Poling** is a [metallurgical] method employed in the purification of copper which contains copper oxide as an impurity and also in the purification of tin ("Sn") which contains tin oxide (stannic oxide or "SnO₂") as an impurity. The impure metal, usually in the form of molten blister copper, is placed in an anode furnace for two stages of refining

6. A mixture of Ca₃P₂ and CaC₂ is used in making Holmes signal. Explain.

Ans:-

- In Holmes signal are used in making the mixture of Ca_3P_2 and CaC_2 .
- This mixture they reacts with water to produce phosphene and acetylene gases which burns and they guides the ships which is in the sea.

CaC_2 .



- In smoke screen are also used the mixture of Ca_3P_2 and CaC_2 .

7. In modern diving apparatus, a mixture of He and O_2 is used.
Why?

Ans: The adding helium is the main reason to the breathing mix the proportions of nitrogen and oxygen below i. e to reduce those of air, the gas mix to be breathed safely on deep dives which they allow. The required to reduce nitrogen narcosis and other physiological effects of the gas at depth for the requirements of lower proportion of nitrogen.

Because -

- In blood helium is less soluble than nitrogen, under high pressure.
- Diving apparatus are used the mixture of oxygen and helium which they used by deep sea divers.

8. Calculate the magnetic moment of a divalent ion in aqueous solution if its atomic number is 25.

Ans:- Atomic number is defined as the number of protons or number of electrons that are present in an atom.

Atomic number = Number of electrons = Number of protons = 25

The electronic configuration is:

The electronic configuration of is, containing 5 unpaired electrons.

Formula used for magnetic moment :

where, μ = magnetic moment

n = number of unpaired electrons

Now put all the given values in the above formula, we get the unpaired electrons.

Thus the magnetic moment will be 5.9 B.M

9. What are artificial sweetening agents? Give example.

Ans:- Artificial sweetening agents are chemicals that sweeten food.

However, unlike natural sweeteners, they do not add calories to our

body. They do not harm the human body. Some artificial sweeteners are aspartame, saccharin, sucrolose, and alitame.

Ex-Sucrolose, saccharin

10. What are antibiotics? Give example.

Ans: An antibiotic is a type of antimicrobial substance active against bacteria and is the most important type of antibacterial agent for fighting bacterial infections. Antibiotic medications are widely used in the treatment and prevention of such infections. They may either kill or inhibit the growth of bacteria.

Ex:-Penicillin

SECTION-B

(6 × 4 = 24)

Note: Answer any six questions:

11. Derive Bragg's equation.

Ans:- :the wavelength. The phases of the beams coincide when the incident angle equals reflecting angle. The rays of the incident beam are in phase and parallel up to point z, which is the point at which top beam strikes the top layer. The second beam passes to next layer and is scattered by B. The second beam travels extra distance $AB + BC$. This extra distance is an integral multiple of

$$n\lambda = AB + BC.$$

But

$$AB = BC$$

$$n\lambda = 2AB \dots (1)$$

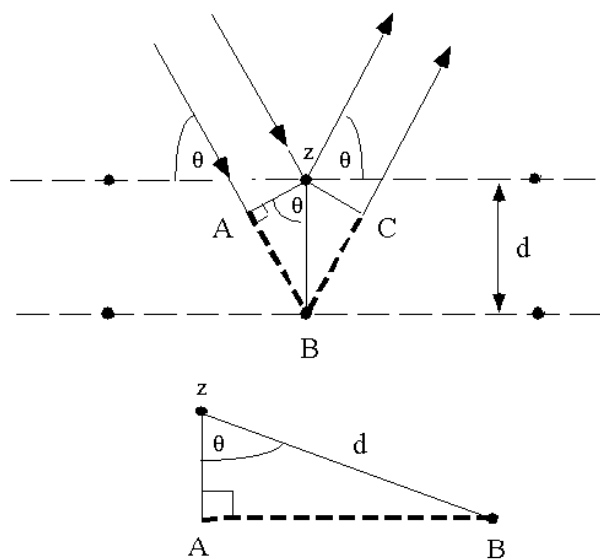
d is the hypotenuse of the right triangle Abz. Ab is opposite to angle θ

$$AB = d \sin \theta \dots (2)$$

Substitute equation (2) in equation (1)

$$n\lambda = 2d \sin \theta$$

This is equation for Bragg's law



12. State Raoult's Law.

The vapour pressure of pure benzene at a certain temperature is 0.850 bar. A non-volatile, non-electrolyte solid weighing 0.5g when added to 39.0g of benzene(molar mass 78 g mol⁻¹). Vapour pressure of the solution then its 0.845 bar. What is the molar mass of the solid substance?

Ans:- Raoult's law is a law of thermodynamics established by French chemist François-Marie Raoult in 1887. It states that the partial pressure of each component of an ideal mixture of liquids is equal to the vapour pressure of the pure component multiplied by its mole fraction in the mixture

The total vapour pressure P of a solution containing two components A and B is

$$p_A = p_A^0 \times x_A$$

$$p_B = p_B^0 \times x_B$$

$$P = p_A + p_B = p_A^0 x_A + p_B^0 x_B$$

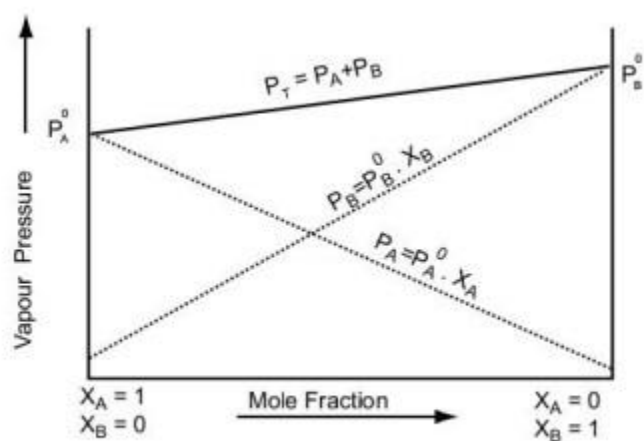
As we know that

$$x_A + x_B = 1$$

$$x_A = (1 - x_B)$$

$$P = (1 - x_B)p_A^0 + p_B^0 x_B$$

$$P = (p_B^0 - p_A^0)x_B + p_A^0$$



Vapour Pressure Diagram for Ideal Solution

$$p^0 = 0.850 \text{ bar}$$

$$p = 0.845 \text{ bar}$$

$$w_B = 0.5 \text{ g}$$

$$w_A = 39.0 \text{ g}$$

$$M_B = ?$$

Applying,

$$\begin{aligned}
 M_B &= \frac{w_B M_A}{w_A \left(\frac{p^0 - p}{p^0} \right)} \\
 &= \frac{0.5 \times 78}{39.0 \times \left(\frac{0.850 - 0.845}{0.850} \right)} \\
 &= \frac{0.5 \times 2 \times 0.850}{0.005} \\
 &= 170 \text{ g mol}^{-1}
 \end{aligned}$$

Hence, molar mass of the solid substance = 170 g/mol

13. What is catalysis? How is catalysis classified? Give two examples for each type of catalysis.

Ans:- A catalyst is a substance that causes a chemical reaction to happen in a different way than it would happen without that catalyst - for example, a catalyst could cause a reaction to happen at a faster rate, or at a lower temperature, than would be possible without the catalyst.

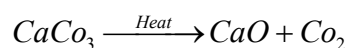
It is classified in two types:-

- i) Homogeneous catalysis-Acid catalysis
- ii) Heterogeneous catalysis-Vanadium oxide

14. Differentiate Roasting and Calcination with examples.

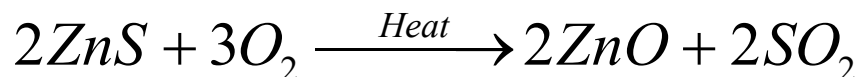
Ans:- Calcination - It is the process of converting an ore into its oxide by heating strongly, below its melting point either in a limited supply or in absence of air. This method is commonly used for converting carbonates and hydroxides to their respective oxides. During calcination, moisture and volatile impurities are also removed.

For example: Calcium carbonate is converted into calcium oxide by calcination.



Roasting - In this process, an ore is converted into its oxide by heating it strongly in excess of air. This method is commonly used for sulphide ores. During roasting, moisture and non metallic impurities are also removed as volatile gases .

For example: Zinc sulphide is converted into zinc oxide by roasting.



The difference between roasting and calcination is that roasting is done in excess of air while calcination is done in limited supply or absence of air.

15. (a) What is Misch metal? Give its composition and uses.

Ans:- Mischmetal is an alloy of rare earth elements with iron and traces of S, C, Ca and

Al The composition is 50% cerium and 25% lanthanum, with small amounts of neodymium and praseodymium. It improves workability of steel. It is used in the ferrocerium "flint" ignition device of many lighters and torches.

It is also used in the preparation of rare earth elements.

(b) What is an Ambidentate Ligand? Give example.

Ans:- A ligand which can bond to the metal ion center through different atoms, but by only one atom at any time, is called ambidentate. Complex. A species made up of a metal ion center bonded to a number of ligand species.

The nitrate ion NO_2^- can bind to the central metal atom/ion at either the nitrogen atom or one of the oxygen atoms.

16. Give the sources of the following Vitamins and name the diseases caused by their deficiency:

a) A b) D c) E d) K

Ans:- Vitamins	Sources	Diseases caused by deficiency
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a)A	Apricots	livedisorders, hematopoiesis
b)D	Cheese, Liver	soft bones, skeletal
c)E	Nuts, Seeds	Disorientation
d)K	Fish, Eggs	Bleeding

17. Explain the terms:

a)Enantiomers b)Racemisation

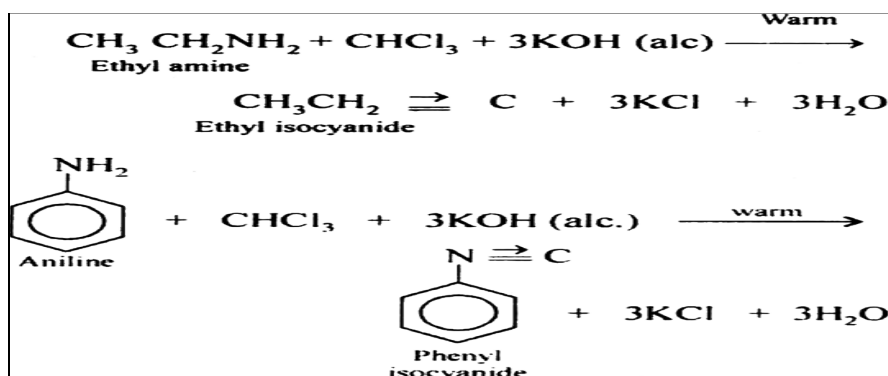
Ans:-(A)Enantiomers:-An enantiomer is one of the two molecules that are mirror images of each other and are non-superposable. Enantiomers have identical chemical and physical properties except for their ability to rotate plane-polarized light (+/-) by equal amounts but in opposite directions. Enantiomers interact differently with other chiral molecules i.e. biologically active molecules as aminoacids, sugars, steroids etc. This means that some molecules have, for example, different odours. Limonene is just such a case.

Ans:-b) Racemisation:-Racemisation is a conversion, by heat or by chemical reaction, of an optically active compound into an optically inactive form which half of the optically active substance becomes its mirror image (enantiomer).

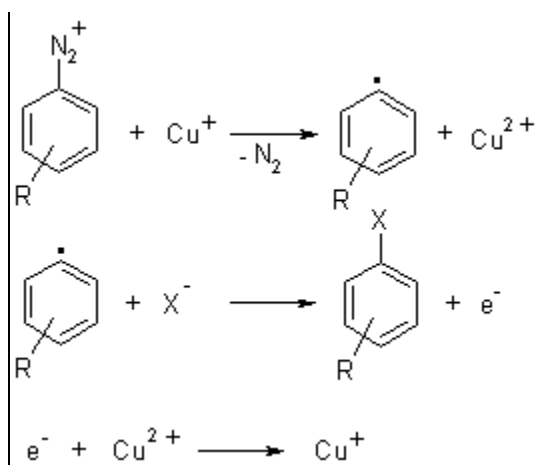
18. Explain the following reactions:

a) Carbylamine reaction b)Sandmeyer reaction

Ans:- a) Primary amine (both aliphatic and aromatic) when warmed with chloroform and alcoholic KOH, gives isocyanides (carbylamines). This is called carbylamine reaction. Carbylamines has an offensive smell. This reaction is answered only by primary amine and hence to distinguish primary amine from other classes of amines.



(B) Sandmeyer reaction:- Sandmeyer reaction is a type of substitution reaction that is widely used in the production of aryl halides from aryl diazonium salts. Copper salts like chloride, bromide, or iodide ions are used as catalysts in this reaction. Notably, Sandmeyer reaction can be used to perform unique transformations on benzene. The transformations include hydroxylation, trifluoromethylation, cyanation, and halogenation.



SECTION-C

(2×8=16)

Note: Answer any two questions:

19. Give a detailed account of the collision theory of reaction rates of Biomolecular gaseous reactions.

Ans:- Collision theory of reaction rates of biomolecular gaseous reaction are:

This theory was given by scientist are Arehenius and von't off.

According to this theory a chemical reaction take place only by collision between the reacting molecules but all collisions are not effective.

only a small fraction of the molecules collide effectively to give the product. To main conditions are there for a collision between the reacting molecules to be productive or effective.

1. The colliding molecules must possess sufficient kinetic energy (threshold energy) to cause a reaction.
2. The reacting molecules must collide with proper orientation.

20.(a) How is chlorine prepared by electrolytic method? Explain its reaction with

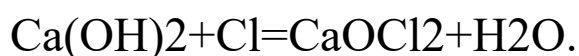
- i) Cold and dil. NaOH ii) Slaked lime

Ans: Chlorine can be manufactured by the electrolysis of a sodium chloride solution (brine), which is known as the Chloralkali process. The production of chlorine results in the co-products caustic soda (sodium hydroxide, NaOH) and hydrogen gas (H₂). These two products, as well as chlorine itself, are highly reactive.

- i) The reaction between chlorine and cold dilute sodium hydroxide solution gives NaClO₃ sodium chlorate(I).



- ii) When Cl₂ reacts with dry slaked lime it forms bleaching powder or chlorine of lime absorbing rapidly.



(b) How does ozone react with following?

- i) PbS ii) Moist KI iii) Hg iv) C₂H₄

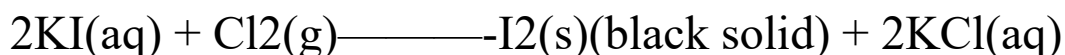
Ans:- i) PbS

Ans:-Lead Sulfide react with ozone to produce lead sulphate and oxygen at room temperature.



ii)KI:-

Ans:-Potassium Iodide reacts with chlorine to give iodine by oxidizing iodide as chlorine is more stronger than KI.



iii)Hg:-

Ans:- Mercury reacts with ozone to form an oxide. Meniscus of mercury is lost and mercurous oxide sticks to the glass. It is known as tailing of mercury.



iv)C₂H₄:-

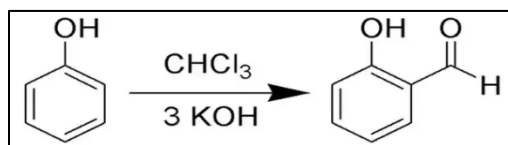
Ans:-Ethylene reacts with ozone to produce ozonide which is decomposed by zn dust and water to form formaldehyde.



21. (a) Explain the following reactions:

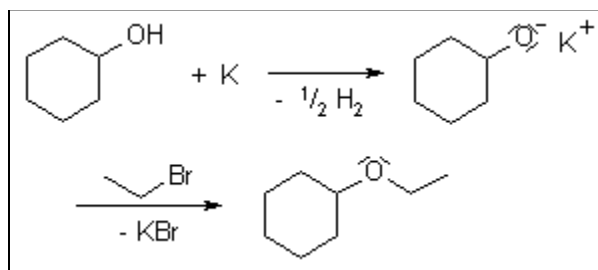
i)Reimer-Tiemann reaction ii)Williamson synthesis

Ans:-i) Reaction of phenol with chloroform and KOH to give salicylaldehyde is known as Reimer-Tiemann reaction.



Ans:- ii) This method is suitable for the preparation of a wide variety of unsymmetric ethers. The nucleophilic substitution of halides with alkoxides leads to the desired products.

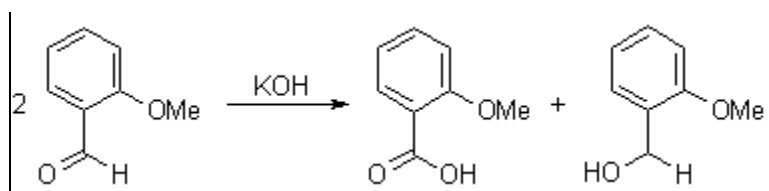
If the halides are sterically demanding and there are accessible protons in the β -position, the alkoxide will act as a base, and side products derived from elimination are isolated instead.



(b) Describe the following:

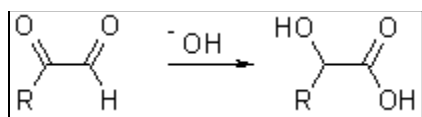
i) Cannizzaro reaction ii) Decarboxylation

Ans:- i) The Cannizzaro reaction, named after its discoverer Stanislao Cannizzaro, is a chemical reaction that involves the base-induced disproportionation of two molecules of a non-enolizable aldehyde to give a primary alcohol and a carboxylic acid.



This redox disproportionation of non-enolizable aldehydes to carboxylic acids and alcohols is conducted in concentrated base.

α -Keto aldehydes give the product of an intramolecular disproportionation in excellent yields.



Ans:- ii Decarboxylation reaction is defined as a chemical reaction that eliminates a carboxyl group and liberates carbon dioxide (CO_2). Decarboxylation mostly refers to a reaction of carboxylic acids erasing a carbon atom from a chain of carbons.

Carboxylation is a completely reversible process which is the first chemical step in photosynthesis, where CO_2 is added to the compound. Whereas, Enzymes that catalyze decarboxylations are known as decarboxylases.

